Hassan Ali Senior Software Engineer

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Summary

Senior software engineer with five years of shipping web products, strongest in TypeScript across NestJS and React. I lead a team of five that took a multi-tenant SaaS from first commit to production, own its AWS infrastructure, and introduced the AI-assisted review workflow the team uses every day. Comfortable at both ends of a project: turning a vague requirement into a phased plan, and owning what happens after it ships. Excited to contribute to innovative projects that prioritize user experience and interface design, leveraging collaboration with cross-functional teams and AI driven enhancements

Professional Experience

Senior Software Engineer, CodingCops 01/2025 – present

Leadership and delivery

- Lead a team of 5 engineers (backend, web, mobile) that took the product from first commit to production: break client requirements into phased technical plans with verification checkpoints, assign the work and review it.
- Reviewed and merged 320+ pull requests; the team has shipped about 2,200 PRs across three codebases.
- Set the engineering standards: clean/hexagonal NestJS architecture, lint + test + build checks on every pull request in Bitbucket Pipelines, pre-commit hooks, and Renovate dependency automation (70+ automated upgrade PRs).
- Introduced an AI-assisted workflow on Claude Code that the team shares: custom agents for PR review against each ticket's acceptance criteria, cross-repo API-contract checks, Prisma migration safety and test writing, plus per-module context docs.

Infrastructure

- Own the AWS infrastructure: ECS Fargate (API and background-worker services), ECR, RDS PostgreSQL, Redis, S3 + CloudFront, SES, CloudWatch and IAM.
- Led the backend's move from EC2 to ECS and the web app's move from Vercel to S3/CloudFront.
- Deploys build a Docker image, register a new ECS task-definition revision, run Prisma migrations as a one-off ECS task, then roll out the new revision.

Product engineering

- Designed the scheduling engine: three auto-scheduling strategies (rotation, fixed days, bi-weekly) with fairness tracking, weekly day-off caps, holidays, blackout dates and swap/vacation/day-off requests.
- Built preview-before-commit for auto-schedules: the real scheduler runs on a BullMQ worker inside a transaction that is rolled back, progress streams over Pusher, and the commit is rejected if the data changed since the preview.
- A Redis lock serializes runs per dealership department; a 250-employee, one-year schedule previews in about 18 seconds.
- Secured a multi-tenant system with 23 roles: JWT access/refresh tokens, email OTP MFA, rate limiting, dealership scoping on every query, role-based impersonation for admin “view as” dashboards.
- Built the web app's scheduling experience in React 19 + Typescript: month calendar with expandable day detail, drag-and-drop shift and vacation moves, conflict warnings and the schedule-preview screen with live progress.
- Added Firebase Analytics (GA4) and Microsoft Clarity to web and mobile, with PII-free route tracking and every build stamped with its version.

Software Engineer, Nathan Digital 07/2023 – 11/2024 | Dubai, UAE

- Collaborated with cross-functional teams to revamp the frontend and backend architecture from traditional to modular component-based structure, improving scalability and modularity by 40%.
- Implemented React-based configurable UI modules for HR workflows, such as leaves, letter generation, and custom approval flows, allowing non-technical users to customize workflows via the UI, reduced client configuration time by 50%.

- Developed 5+ workflows related private packages using TypeScript, enhanced platform functionality and significantly reduced development time.
- Enforced secure authentication mechanisms using JSON Web Tokens (JWT), and integrated Single Sign-On (SSO) solutions, including Microsoft SSO, ensuring seamless and secure access across different services.
- Integrated GPT models into client applications, enabling AI-driven features such as automated support and data-driven content generation, enhancing user engagement.

JavaScript Developer, Reown Logics 10/2021 – 07/2023 | Lahore, Pakistan

- Crafted landing pages for over 5 branding websites, incorporated interactive animations and SEO best practices ensuring application performance and responsiveness.
- Built a video recording web application to record/upload and share 60 seconds business proposal video using Media Recorder API, React, NodeJS, MongoDB and S3.
- Engineered the frontend of a lead generation platform for job postings using React.
- Collaborated on integrating basic GPT models for content generation in job postings and candidate matching, streamlining recruitment processes and improving time to hire by 30%.
- Built workflows and integrated webhooks, Integration of 3rd party APIs to automate leads generation from different CRMs to one single dashboard and reduced cost by more than 50%.

Skills

- JavaScript, TypeScript
- MUI, Tailwind
- AWS
- MongoDB
- Git, Github, Bit Bucket
- React
- Node.js, NestJS
- OpenAI API
- Jira
- Heroku, cPanel
- Redux, React Query
- RESTful API
- Postgres
- Web Sockets, Pushers
- Cursor AI

Education

Bachelors in Computer Engineering, The University of Lahore

Passion and Hobbies

Chess
Avid chess player, leveraging strategic thinking in problem-solving.

LLM & AI
Exploring LLM along with RAG for meaningful solutions and automate complex workflows.

References

References available upon request.